

TAI NGUYEN XUAN

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EDUCATION

- **University of Science and Technology of Hanoi (USTH)** Oct 2022 – Aug 2026
Bachelor in Advanced Materials Science and Nanotechnology (AMSN)
 - GPA: 17.01/20 | Ranked 1st in Major (Valedictorian)
 - Thesis: "Fabrication and Characterization of Polydopamine-based Crossbar Devices"

RESEARCH EXPERIENCE

Laboratory of Advanced Materials and Application

University of Science and Technology of Hanoi

- **Fabrication and Characterization of Polydopamine-based Crossbar Devices** Sep 2025 – Present
Research Assistant | Supervisor: Dr. Nguyen Van Quyen
 - Fabricated polydopamine (PDA) crossbar memory junctions by integrating electrodeposited thin films with electron-beam evaporated metal electrodes.
 - Performed DC I-V and temperature-dependent electrical characterization to investigate switching behavior and charge transport, supported by AFM, Raman, and UV-Vis analysis.
 - Developed LabVIEW-based automated measurement routines for fast-pulse switching, endurance, retention, and switching dynamics.
- **Visiting Research Scholar** June 2026
Molecular Electronics Lab, Guangdong Technion Israel Institute of Technology, China
 - Quantified short-term plasticity (STP), paired-pulse facilitation (PPF), and endurance to evaluate the temporal response and cycling stability of volatile PDA memristors.
 - Conducted XPS analysis of PDA thin films to examine surface composition and chemical states relevant to protonation and interfacial transport.
- **Design and Fabrication of a Low-Temperature Cryostat System** Nov 2025 – Mar 2026
Project Group Leader | Supervisor: Dr. Nguyen Van Quyen
 - Designed and integrated a modular low-temperature electrical characterization system with cryogenic-compatible components and custom PCBs.
 - Modeled transient heat transfer using Python-based finite-difference simulations to guide thermal design, fabrication, and system integration.
- **Simulation of Plasmonic Structures** Sep 2024 – Jun 2025
Research Assistant | Supervisors: Dr. Hoang Thi Hong Cam, Assoc. Prof. Nguyen Van Quynh
 - Modeled Au/Au@Ag plasmonic nanosphere arrays using Lumerical FDTD and HEAT to investigate optical resonance and associated thermal transport.

PUBLICATIONS & MANUSCRIPTS

- **Nguyen, T. X., Le, T., Tran, V. T., Nguyen, Q.,** "Electric Field-Driven Interfacial Proton-Coupled Electron Transfer in Protonated Polydopamine Junctions." *Submitted to Journal of the American Chemical Society*, 2026.

ADDITIONAL TRAINING

- **Photovoltaic Summer School, Tatung University, Taiwan** Jun 2025
Coursework and laboratory training in photovoltaic and smart-energy systems.

HONORS AND AWARDS

- **Merit Scholarship, USTH** 2023-2025
Selective merit scholarship awarded for academic excellence.
- **Shaping the Future Initiative Scholarship, USTH** 2025-2026
Awarded to select students for exceptional academic and research achievements.
- **Rector's Award of Excellence, USTH** 2022-2025
Highly selective annual award presented to only 10 students university-wide for remarkable contributions to student leadership and institutional development.
- **Internship Scholarship, USTH** 2025-2026
Competitive funding for high-potential undergraduate research internships.

SKILLS

- **Fabrication:** Photolithography, e-beam evaporation, electrodeposition, thin-film processing
- **Materials Characterization:** XPS, AFM, Raman/SERS, UV-Vis, FTIR, optical microscopy
- **Electrical Characterization:** DC I-V, temperature-dependent I-V, fast-pulse characterization, endurance/retention
- **Instrumentation & Automation:** LabVIEW, instrument control, automated measurement workflows
- **Simulation & Engineering Design:** Lumerical FDTD/HEAT, COMSOL, finite-difference modeling, FreeCAD, KiCad
- **Programming & Research Computing:** Python (NumPy, pandas, Matplotlib), Git/GitHub, reproducible scientific workflows
- **Scientific Documentation:** Typst, $\LaTeX$
- **Languages:** Vietnamese (native), English (IELTS 6.5), French (B1)

TEACHING & LEADERSHIP EXPERIENCE

- **Academic Teaching & Mentoring** Sep 2023 – Mar 2026
Tutor & Peer Mentor
 - Tutored undergraduate students in fundamental physics and chemistry, providing structured academic support.
- **USTH Science Art Club** Jul 2024 – Feb 2025
Club President
 - Led STEM outreach and taught physics and chemistry to visiting high-school students.
 - Trained members in scientific literature analysis and coordinated STEM exhibition projects.
- **USTH Student Association** Mar 2023 – Jun 2024
Marketing Committee Member
 - Coordinated campus-wide student initiatives and communication with university offices.
 - Delivered a time-management workshop for incoming students.